



PAPER – 3: QUANTITATIVE APTITUDE



QUESTIONS

1. If arithmetic mean between roots of a quadratic equation is 8 and the geometric mean between them is 5, the equation is _____
 - (a) $x^2 - 16x - 25 = 0$
 - (b) $x^2 - 16x + 25 = 0$
 - (c) $x^2 - 16x + 25 = 0$
 - (d) None of these
2. Roots of equation $2x^2 + 3x + 7 = 0$ are α and β . The value of $\alpha\beta^{-1} + \beta\alpha^{-1}$ is
 - (a) 2
 - (b) $3/7$
 - (c) $7/2$
 - (d) $-19/14$
3. If $\frac{p}{q} = -\frac{2}{3}$ then the value of $\frac{2p+q}{2p-q}$ is:
 - (a) 1
 - (b) $-1/7$
 - (c) $1/7$
 - (d) 7

4. Find the value of $[\log_{10}\sqrt{25} - \log_{10}(2^3) + \log_{10}(4)^2]^x$
- (a) x
 - (b) 10
 - (c) 1
 - (d) None
5. A sum of money doubles itself in 10 years. The number of years it would treble itself is:
- (a) 25 years
 - (b) 15 years
 - (c) 20 years
 - (d) None
6. The effective rate equivalent to nominal rate of 6% compounded monthly is:
- (a) 6.05
 - (b) 6.16
 - (c) 6.26
 - (d) 6.07
7. What is the rate of simple interest if a sum of money amounts to Rs. 2,784 in 4 years and Rs. 2,688 in 3 years?
- (a) 1% p.a.
 - (b) 4% p.a.
 - (c) 5% p.a.
 - (d) 8% p.a.
8. A building contractor needs three helpers and ten men apply. In how many ways can these selections take place?
- (a) 36
 - (b) 15

- (c) 150
(d) 120
9. An examination paper consists of 12 questions divided into two parts A and B. Part A contains 7 questions and Part B contains 5 questions. A candidate is required to attempt 8 questions selecting at least 3 from each part, in how many maximum ways can the candidate select the questions?
- (a) 35
(b) 175
(c) 210
(d) 420
10. If $A = (1,2,3,4,5)$, $B = (2,4)$ and $C = (1,3,5)$ then $(A-C) \times B$ is
- (a) $\{(2,2), (2,4), (4,2), (4,4), (5,2), (5,4)\}$
(b) $\{(1,2), (1,4), (3,2), (3,4), (5,2), (5,4)\}$
(c) $(2,2), (4,2), (4,4), (4,5)$
(d) $(2,2), (2,4), (4,2), (4,4)$
11. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function, defined by $f(x) = 10x-7$, if $g(x) = f^{-1}(x)$, then $g(x)$ is equal to
- (a) $\frac{1}{10x-7}$
(b) $\frac{1}{10x+7}$
(c) $\frac{x+7}{10}$
(d) $\frac{x-7}{10}$

12. $\lim_{n \rightarrow \infty} \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n} \right)$ is equal to :
- (a) $\frac{1}{2}$
- (b) $\frac{1}{3}$
- (c) 2
- (d) 1
13. The function $f(x) = \frac{x^2 - 9}{x - 3}$ is undefined at $x = 3$. What value must be assigned to $f(3)$, if $f(x)$ is to be continuous at $x = 3$?
- (a) 6
- (b) 0
- (c) 9
- (d) 3
14. Given $x = 2t + 5$; $y = t^2 - 2$, then $\frac{dy}{dx}$ is calculated as:
- (a) t
- (b) $1/t$
- (c) $-1/t$
- (d) None
15. $\int_1^2 \frac{2x}{1+x^2} dx$:
- (a) $\log_e \frac{5}{2}$
- (b) $\log_e 5 - \log_e 2 + 1$
- (c) $\log_e \frac{2}{5}$

- (d) None of these
16. In certain code language 'CLOCK' is coded as 75276 and 'EARTH' is coded as 83491, then 'COAT' is coded as
- (a) 7329
(b) 7239
(c) 7932
(d) 7529
17. Find the missing term of series 2, 7, 16, 29...., 67, 92
- (a) 39
(b) 46
(c) 43
(d) 62
18. In a certain language 'MENTION' is written as 'NFOUJPO', the code of 'MYSTIFY' is:
- (a) NZTUJGZ
(b) NFOFTJT
(c) LNEITNO
(d) OERESTIN
19. Anil started walking 5 kms towards north then he turned left and walked 3 kms. Again, he turned left and walked 5 kms. Then the total number of kms he walked is
- (a) 13 kms
(b) 8 kms
(c) 3 kms
(d) 5 kms

20. Raju started walking 10 kms towards east from his home. He turned right and walked 5 kms to the south to reach his school. In which directions is his school from his home?
- (a) South – East
 - (b) North – East
 - (c) South – West
 - (d) North – West
21. L is wife of N, P is son of N, K is brother of N and father of O. What is the relationship of P and O?
- (a) Uncle
 - (b) Brother
 - (c) Cousin
 - (d) Nephew
22. Standard Error (SE) and square root of sample size are
- (a) Directly proportional
 - (b) Equal
 - (c) Inversely proportional
 - (d) Not equal
23. Out of 1000 persons 40% are female, others are male. In a marriage function, 300 persons enjoyed the song. 30% of the people who had not enjoyed the song were female. What is the number of male, who did not enjoy the song in the function?
- (a) 120
 - (b) 180
 - (c) 360
 - (d) 490
24. Find the Harmonic Mean of 2, 4 & 6.
- (a) 3.30
 - (b) 3.00

- (c) 3.75
 - (d) 4.00
25. If the mode of the following data is 13, then the value of x in the data set is 13, 8, 6, 3, 8, $13, 2x + 3$, 8, 13, 3, 5, 7
- (a) 6
 - (b) 5
 - (c) 7
 - (d) 8
26. The best measure of central tendency is
- (a) Mean
 - (b) Median
 - (c) Mode
 - (d) Range
27. For a distribution the mean is 30. The standard deviation is 2, then coefficient of variation is.
- (a) 6.67%
 - (b) 9.45%
 - (c) 7.5%
 - (d) 2.5%
28. Mean deviation is ____ when the deviations are taken from the median.
- (a) maximum
 - (b) minimum
 - (c) zero
 - (d) can't say
29. Ogive is used to find
- (a) Mean
 - (b) Median
 - (c) Mode

- (d) Range
30. A population comprises 7 members. The number of all possible samples of size 3 that can be drawn from it with replacement is
- (a) 216
 - (b) 343
 - (c) 21
 - (d) 125